

Exploring Cognitive Performance Across Education Levels and Alzheimer's Groups



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Introduction

Motivation: I chose this topic because I am interested in how an individual's environment and experiences (such as years in education) impacts their ability to make decisions, which is a cognitive function that is disrupted by Alzheimer's disease.

Research Questions:

Research Question 1 – Are MMSE scores associated with years of education among older adults with and without Alzheimer's disease?

Research Question 2 – Do MMSE scores differ between individuals diagnosed with Alzheimer's disease and those who aren't?

Methods

Data Collection:

- Each sample unit represents one person from the OASIS dataset
- My final sample size was 100 participants. I removed all returning visit data and participants listed as “Converting.” From this sample, I randomly selected 50 of each diagnostic group.

Measures:

- MMSE Score: cognitive test with scores ranging 0-30 points
- Years of Education: total years of schooling ranging 6-24 years
- Diagnostic Group: clinical group for with Alzheimer’s (“Demented”) and without (“Nondemented”)

Analysis Method: I ran a GLM (Multiple Regression) in RStudio.

Descriptives

Table 1 – Descriptive Statistics

	<i>Median</i>	<i>IQR</i>
<i>MMSE Score (MMSE)</i>	29 points	3.5 points
	<i>Median</i>	<i>IQR</i>
<i>Years of Education (EDUC)</i>	14 years	4 years
	<i>Demented</i>	<i>Nondemented</i>
<i>Diagnostic Group (Group)</i>	50	50

Results

Results table:

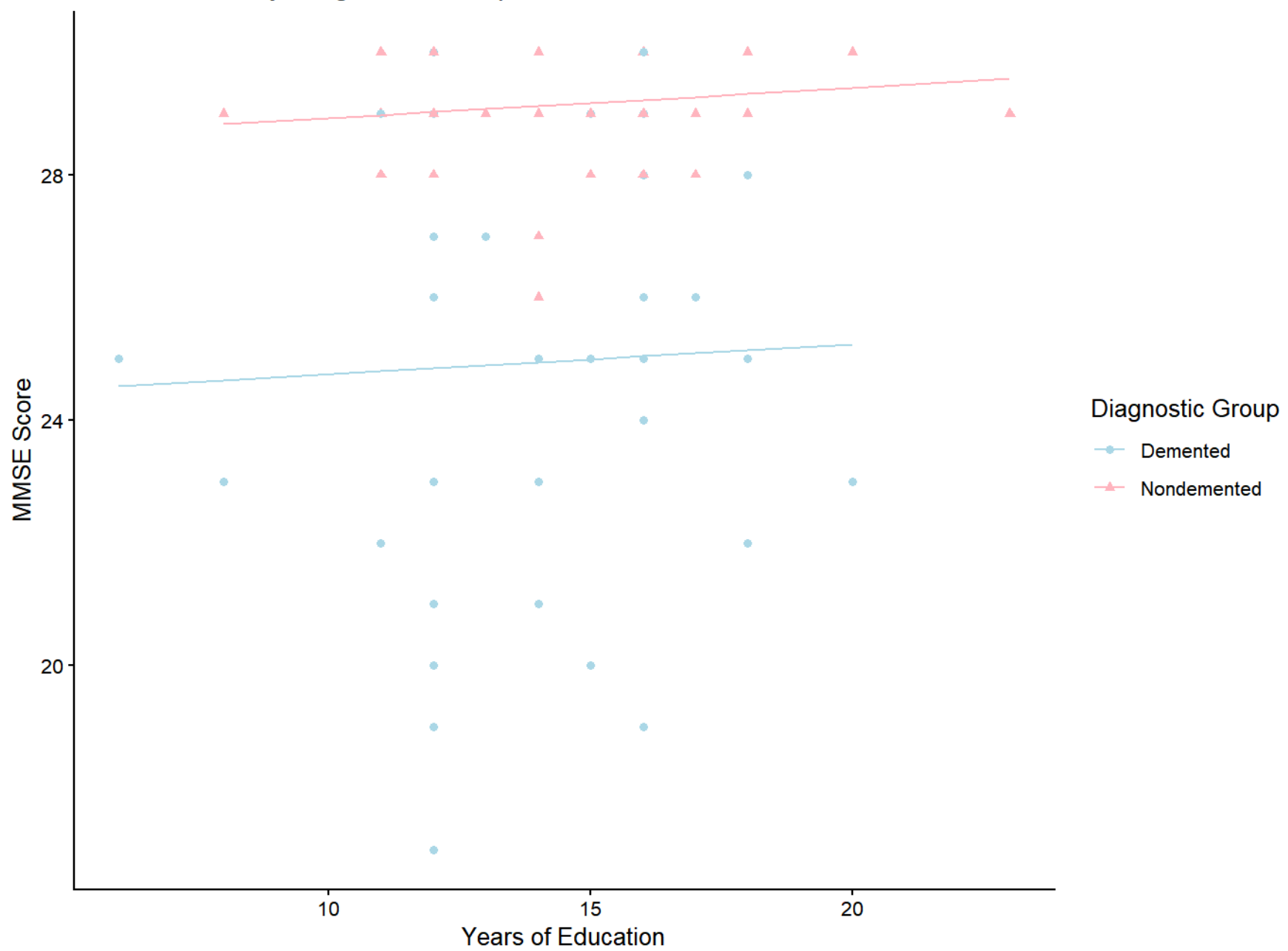
Table 2 – Model Results

	Estimate	St. Error	t-stat	p-value
Intercept	24.259	1.275	19.03	<2e-16
EDUC	0.049	0.091	0.54	0.591
GroupNondemented	4.177	0.518	8.06	2e-12

Model df: (2, 97)

Adjusted R^2 : 0.4226

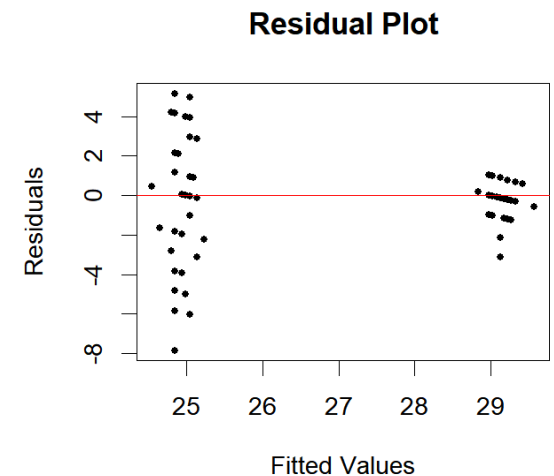
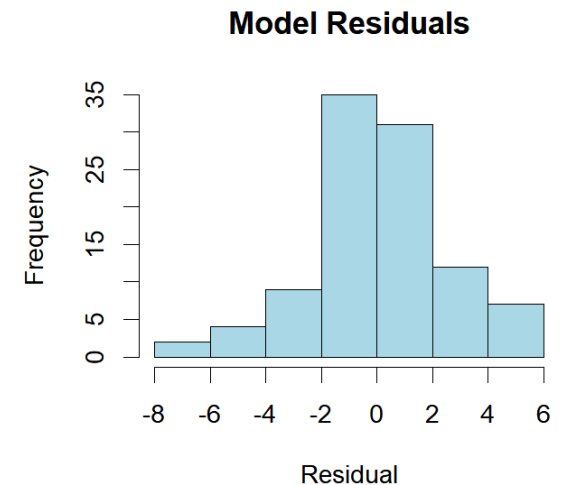
MMSE Score by Diagnostic Group and Years of Education



Assumptions

Assumptions:

- Random sample: met – I randomly selected 100 people from the study
- Independent observations: met – One person's MMSE score is unrelated to another person's MMSE score
- Years of Education is linearly related to MMSE score: met – no curves found in the residual plot
- Residuals are normally distributed: met – model residuals histogram shows normal distribution
- Residuals have equal variance: not met – residuals show decreasing trend left to right



Discussion

Interpretation:

- When controlling for Diagnosis Group, Years of Education is not significantly, linearly related to MMSE score ($t=0.54$, $df=97$, $p>0.05$)
- When controlling for Years of Education, people with Alzheimer's and people without Alzheimer's significantly differ in MMSE Score ($t=8.06$, $df=97$, $p>0.05$)
 - 'Demented' patients had significantly lower MMSE scores than 'Nondemented' patients, when controlling for Years of Education

Limitations: The residuals did not have equal variance, as funneling was observed. Additionally, while I did randomly select 50 participants per diagnosis, the original study collected participants from a database of previous subjects in MRI studies at WUSTL.

Implications: Overall, individuals who did not have Alzheimer's had higher MMSE cognitive test scores. However, the years of education obtained had no significant relationship with MMSE score.

Future Research: In the future, I would include other variables, such as time since diagnosis, previous neurological or vascular issues (stroke, heart attack, TBI), median income, or environmental living conditions, as these all might relate to the MMSE score. I would be most interested in median income, as I would hypothesize that individuals with more money may be able to afford access to more or better education, food, and resources, therefore enhancing their physical and cognitive abilities.

References:

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